

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

NAME (USN)

**in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING**



**B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
February to June 2026**

CERTIFICATE

**B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering**



This is to certify that the Lab work entitled “**ANALYSIS AND DESIGN OF ALGORITHMS**” carried out by NAME (USN), who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Index Sheet

Sl. No.	Experiment Title	Page No.
1	Write program to obtain the Topological ordering of vertices in a given digraph.	
2	Implement Johnson Trotter algorithm to generate permutations.	
3	Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.	
4	Sort a given set of N integer elements using Quick Sort technique and compute its time taken.	
5	Sort a given set of N integer elements using Heap Sort technique and compute its time taken.	
6	Implement 0/1 Knapsack problem using dynamic programming.	
7	Implement All Pair Shortest paths problem using Floyd's algorithm.	
8	(a) Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm. (b) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.	
9	Implement Fractional Knapsack using Greedy technique.	
10	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.	
11	Implement "N-Queens Problem" using Backtracking.	

Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1:

Complete program in C

Screenshot of the output to be included for all the cases.

For all sorting programs, graph (plotted in Excel) to be attached and shown for large values of 'n' how time increases.

**NOTE: SIMILARLY INCLUDE FOR ALL THE LAB PROGRAMS (ALL 11).
LEETCODE PROGRAMS ALSO TO BE INCLUDED. Include the Leetcode Programs
which you have executed along with the respective lab program.**

ENSURE THAT EACH LAB PROGRAM STARTS IN A NEW PAGE ONLY.